

Computational Physics

Summer 2026

Lecture #20

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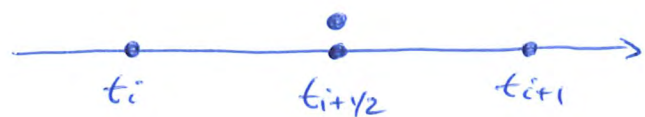
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Adaptive Stepsize - Approach #1

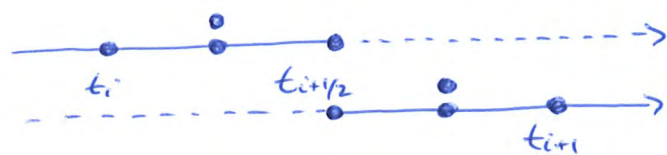
①

- idea:
- (1) do every step twice, once with (h) , once with $(h/2)$
 - (2) compare solutions
 - (3) if difference $\epsilon > \epsilon_{tol}$, reduce h and repeat step

Ex: RK4



4 evaluations (h)



8-1=7 evaluations $(h/2) \times 2$

Compare the two results:

$$\epsilon = |y_{i+1}^{(h)} - y_{i+1}^{(h/2)}| \approx \tau_{i+1} \text{ is an estimate of the local truncation error}$$

$\uparrow_h \quad \uparrow_{h/2} \quad \propto h^5$

requiring $\epsilon = \epsilon_{tol} \Rightarrow \frac{\epsilon_{tol}}{\epsilon} = \left(\frac{h_{tol}}{h}\right)^5 \Rightarrow \boxed{h_{tol} = h \left(\frac{\epsilon_{tol}}{\epsilon}\right)^{1/5}}$

- if $\epsilon > \epsilon_{tol}$, repeat step with $h_{new} = h_{tol}$
- if $\epsilon \approx \epsilon_{tol}$, continue with h
- if $\epsilon \ll \epsilon_{tol}$, can adopt larger $h_{new} = 2h$

Approach #2

- idea: apply two RK schemes of different orders, same stepsize (h)
- common intermediate nodes $\{p, p-1\}$
- $\Rightarrow$ inexpensive error estimate, enables adaptive stepsize selection

second method: $y_{i+1} = y_i + h \sum_{j=1}^n b_j' k_j$, $k_j' = k_j$

$\Rightarrow \tau_{i+1} = y_{i+1} - y_{i+1}' = h \sum_{j=1}^n (b_j - b_j') k_j$ is $\mathcal{O}(h^p)$

Ex: forward Euler ($p=1$) and modified Euler RK2 ($p=2$)

0	
1	1
	$\frac{1}{2} \quad \frac{1}{2}$
	$1 \quad 0$

"extended tableau"

$\left. \begin{array}{l} \text{two weight rows give} \\ b_j, b_j' \text{ of the two methods} \end{array} \right\} \Rightarrow \epsilon$

Ex: Runge-Kutta-Fehlberg method (RK45)

(2)

- $\mathcal{O}(h^4)$ scheme with $\mathcal{O}(h^5)$ error estimate
- requires only one extra evaluation!

0					
$\frac{1}{4}$	$\frac{1}{4}$				
$\frac{3}{8}$	$\frac{3}{32}$	$\frac{9}{32}$			
$\frac{12}{13}$	$\frac{1932}{2197}$	$-\frac{7200}{2197}$	$\frac{7296}{2197}$		
1	$\frac{439}{216}$	-8	$\frac{3680}{513}$	$-\frac{845}{4104}$	
$\frac{1}{2}$	$-\frac{8}{27}$	2	$-\frac{3544}{2565}$	$\frac{1859}{4104}$	$-\frac{11}{40}$
	$\frac{25}{216}$	0	$\frac{1408}{2565}$	$\frac{2197}{4104}$	$-\frac{1}{5}$
	$\frac{16}{135}$	0	$\frac{6656}{12825}$	$\frac{28561}{56470}$	$-\frac{9}{50}$

$$y_{i+1} = y_i + h \left[\frac{25}{216} k_1 + \dots - \frac{1}{5} k_5 \right]$$

$$y'_{i+1} = y'_i + h \left[\frac{16}{135} k_1 + \dots + \frac{2}{55} k_6 \right]$$

$$\Rightarrow E = |y_{i+1} - y'_{i+1}| = h \left[\frac{1}{360} k_1 + \frac{1}{33} k_3 - \frac{1}{34} k_4 + \frac{1}{50} k_5 + \frac{2}{55} k_6 \right]$$

$$\Rightarrow h_{tol} = 0.84 h \left(\frac{E_{tol}}{E} \right)^{1/5}$$

expensive to repeat steps, so be conservative

note: RK45 advanced with y_{i+1} not y'_{i+1} , still $\mathcal{O}(h^4)$ since we need error

Other Methods

■ m-step multistep methods

(1) Adams-Bashforth

explicit $\mathcal{O}(h^4)$ $m=4$

(2) Adams-Moulton

implicit $\mathcal{O}(h^4)$ $m=3$

(3) predictor-corrector: explicit forward 'prediction', then implicit 'correction'

- Euler + trapezoidal, can iterate until $E < E_{tol}$

- Adams-Bashforth-Moulton

(4) Bulirsch-Stoer

- a series of decreasing stepsizes h_j , then extrapolate to $\lim h \rightarrow 0$

■ Leap-frog $\left[\frac{d^2 y}{dx^2} = A(y) \right]$

■ Verlet

■ Numerov method $\left[\text{Sturm-Liouville problems} \right]$

e.g. time-independent Schrödinger equation

$$\frac{d}{dx} \left[p(x) \frac{dy(x)}{dx} \right] + q(x) y(x) = \lambda w(x) y(x)$$

where $p(x), q(x), w(x)$ are 'weight functions'

λ are eigenvalues

Sturm-Liouville-type equations

(3)

- 2nd order ODEs with no 1st order derivatives

$$\boxed{\frac{d}{dx} \left[p(x) \frac{dy(x)}{dx} \right] + q(x)y(x) = \lambda w(x)y(x)}$$

boundary value problems: specify $y(a) = y_a$, $y(b) = y_b$

Ex: Laplace equation

$$y''(x) = 0 \quad \nabla^2 y = 0$$

↑ if we have a "source term" on the RHS $\Rightarrow$ Poisson's equation

- seen in E&M (electrostatics), gravitation, fluid dynamics, etc

Ex: Legendre equation

$$(1-x^2)y'' - 2xy' + v(v+1)y = 0$$

- seen e.g. in the radial potential for Laplace in spherical coordinates

- solutions are the Legendre polynomials

! has a first derivative!

↳ can rewrite as $[(1-x^2)y'(x)]' + v(v+1)y(x) = 0$

$$\left[\begin{aligned} \text{since } [] &= [y'(x) - x^2 y'(x)]' = y''(x) - \frac{d}{dx} [x^2 y'(x)] \\ &= y''(x) - \frac{dx^2}{dx} y'(x) - x^2 y''(x) = (1-x^2)y''(x) - 2xy'(x) \end{aligned} \right]$$

Thm: in general, can transform

$P(x)y''(x) + Q(x)y'(x) + R(x)y(x) = 0$ into SL form

i.e. eliminate $y'(x)$ term by multiplying by $m(x) = \frac{1}{P(x)} \exp \left[\int \frac{Q(x)}{P(x)} dx \right]$

Thm: SL form (above) can be written as an eigenvalue problem

$$\boxed{\mathcal{L}y(x) = \lambda y(x)}$$

where $\mathcal{L} = \frac{1}{w(x)} \left[\frac{d}{dx} \left(p(x) \frac{d}{dx} \right) + q(x) \right]$ the "SL operator"

is a linear differential operator

Thm: $\mathcal{L}$ is Hermitian $\Rightarrow L_{ij} = \overline{L_{ji}}$ where $\overline{a+ib} = a-ib$ "complex conjugate"

- the solutions $y(x)$ are an orthogonal set of basis eigenfunctions ϕ_i , with real eigenvalues λ_i

$\Rightarrow$ can define matrix $L_{ij} = \int \bar{\phi}_i(x) \mathcal{L} \phi_j(x) dx$ [future lectures: linear algebra approach]

Schrödinger Equation

- important SL-type problem in QM

full, time-dependent: $i\hbar \frac{\partial}{\partial t} \Psi(\vec{x}, t) = \hat{H} \Psi(\vec{x}, t)$

where $\Psi(\vec{x}, t)$ is the wave function

$|\Psi|^2 = \mathcal{P}$ is the probability to find a particle at $(\vec{x}, t)$

$\hat{H}$ is the Hamiltonian

- ansatz: solution has separable space and time parts

$$\Rightarrow \Psi(\vec{x}, t) = e^{-iEt} \psi(\vec{x})$$

gives the time-independent ("stationary") equation for the spatial part:

$$\hat{H} \psi(\vec{x}) = E \psi(\vec{x})$$

is an eigenvalue problem, with $\lambda = \text{energy } E$ of the Hamiltonian

Ex: consider $\hat{H} = \underbrace{-\frac{\hbar^2}{2m} \nabla^2}_{KE} + \underbrace{V(\vec{x})}_{PE}$

$$\begin{aligned} &= \hat{E}_{kin} + \hat{E}_{pot} \\ &= \frac{\vec{p}^2}{2m} + V(\vec{x}) \\ &= \frac{-\hbar^2}{2m} \nabla^2 + V(\vec{x}) \\ &\left[\text{since } \vec{p} = -i\hbar \nabla \right] \end{aligned}$$

in the one-dimensional case:

$$\boxed{\psi''(x) + \underbrace{\frac{2m}{\hbar^2} [E - V(x)]}_{g(x)} \psi(x) = 0}$$

$\Rightarrow \psi''(x) + g(x)\psi(x) = 0$ is SL-like in form

note: in 3D, $\Psi(r, \theta, \phi) = R(r) Y_{lm}(\theta, \phi)$

where Y_{lm} are the spherical harmonics, also gives a SL-like equation

$$\frac{d^2 Y_l(r)}{dr^2} + \frac{2m}{\hbar^2} \left[E - V(r) - \frac{\hbar^2 l(l+1)}{2mr^2} \right] Y_l(r) = 0$$

Numerov Method

(5)

- $\mathcal{O}(h^6)$ one-step solver for SL-type equations

idea: use Taylor expansion, eliminate high order terms due to symmetries

$$\text{forward: } y_{i+1} = y_i + hy_i' + \frac{h^2}{2!} y_i'' + \frac{h^3}{3!} y_i^{(3)} + \frac{h^4}{4!} y_i^{(4)} + \mathcal{O}(h^5)$$

$$\text{backward: } y_{i-1} = y_i - hy_i' + \frac{h^2}{2!} y_i'' - \frac{h^3}{3!} y_i^{(3)} + \frac{h^4}{4!} y_i^{(4)} + \mathcal{O}(h^5)$$

$$\text{sum together: } y_{i+1} + y_{i-1} = 2y_i + h^2 y_i'' + \frac{h^4}{12} y_i^{(4)} + \mathcal{O}(h^6) \quad \begin{array}{l} \text{odd terms} \\ \text{drop out} \end{array}$$

↳ use this expression twice:

#1 - general numerical estimate of 2nd derivative, $y''(t)$ at t_i

$$\Rightarrow \boxed{\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} = y_i'' + \mathcal{O}(h^2)} \quad \text{"centered difference"}$$

↑ accurate / with leading error term $\mathcal{O}(h^2)$

#2 - keeping the $y_i^{(4)}$ term, to derive our scheme:

$$y_{i+1} + y_{i-1} = 2y_i + h^2 y_i'' + \frac{h^4}{12} y_i^{(4)} + \mathcal{O}(h^6) \quad (*)$$

↳ $\frac{h^4}{12}$

Schrödinger $\square$: $\psi''(x) = -g(x)\psi(x)$ gives us the second derivative

notation: $\psi \rightarrow y$, $y_i'' = -g_i y_i$

? : differentiate twice to get 4th derivative, use approximation above:

$$y_i^{(4)} = [-g_i y_i]'' = \frac{-g_{i+1} y_{i+1} + 2g_i y_i - g_{i-1} y_{i-1}}{h^2}$$

substitute into (*):

$$y_{i+1} + y_{i-1} = 2y_i - h^2 g_i y_i + \frac{h^2}{12} [-g_{i+1} y_{i+1} + 2g_i y_i - g_{i-1} y_{i-1}]$$

solve for y_{i+1} to obtain the update rule:

$$\boxed{\left(1 + \frac{h^2}{12} g_{i+1}\right) y_{i+1} = \left(2 - \frac{5h^2}{6} g_i\right) y_i - \left(1 + \frac{h^2}{12} g_{i-1}\right) y_{i-1} + \mathcal{O}(h^4)}$$

- with just two function evaluations $\{y_i, y_{i-1}\}$ we can advance (h) with $\tau_{\text{local}} \sim \mathcal{O}(h^6)$

note: just like leap-frog, need to 'initialize' at the very beginning

↳ use some other / simpler scheme e.g. RK